

GML_06_01-Case-Toyota.R

frank

2024-05-06

```
# Author Frank Lehrbass, May 2022
# Ein R Skript für das FOM Zertifikat
# nur zum Zwecke der Lehre
# KEINE GARANTIE bzgl Korrektheit o.a.
# This R coding is WITHOUT ANY WARRANTY

#Zuvor verwendete Variablen Leeren

# Clean previous variables
rm(list=ls(all=TRUE))

# Load packages

library(lmtest) # Regressionsdiagnostik

## Warning: Paket 'lmtest' wurde unter R Version 4.3.3 erstellt

## Lade nötiges Paket: zoo

##
## Attache Paket: 'zoo'

## Die folgenden Objekte sind maskiert von 'package:base':
##
##      as.Date, as.Date.numeric

library(sandwich) #HAC SE

## Warning: Paket 'sandwich' wurde unter R Version 4.3.3 erstellt

library(car) #VIF und Co

## Warning: Paket 'car' wurde unter R Version 4.3.3 erstellt

## Lade nötiges Paket: carData

library(sjPlot) #only needed for nice output/fomattting of tables see viewer
#if THIS PACKAGE is not available it is no big issue, content stays the same,
just layout less nice

Daten = read.table("GML_06_01 Case Toyota.csv", header=TRUE, sep=";", dec=".")

summary(Daten)
```

```

##      Id      Model      Price      Age_08_04
##  Min.   : 1.0    Length:1436    Min.   : 4350    Min.   : 1.00
## 1st Qu.: 361.8    Class :character    1st Qu.: 8450    1st Qu.:44.00
## Median : 721.5    Mode  :character    Median : 9900    Median :61.00
## Mean   : 721.6                    Mean  :10731    Mean  :55.95
## 3rd Qu.:1081.2                    3rd Qu.:11950    3rd Qu.:70.00
## Max.   :1442.0                    Max.   :32500    Max.   :80.00
##      Mfg_Month      Mfg_Year      KM      Fuel_Type
##  Min.   : 1.000    Min.   :1998    Min.   : 1    Length:1436
## 1st Qu.: 3.000    1st Qu.:1998    1st Qu.: 43000    Class :character
## Median : 5.000    Median :1999    Median : 63390    Mode  :character
## Mean   : 5.549    Mean  :2000    Mean  : 68533
## 3rd Qu.: 8.000    3rd Qu.:2001    3rd Qu.: 87021
## Max.   :12.000    Max.   :2004    Max.   :243000
##      HP      Met_Color      Color      Automatic
##  Min.   : 69.0    Min.   :0.0000    Length:1436    Min.   :0.00000
## 1st Qu.: 90.0    1st Qu.:0.0000    Class :character    1st Qu.:0.00000
## Median :110.0    Median :1.0000    Mode  :character    Median :0.00000
## Mean   :101.5    Mean  :0.6748                    Mean  :0.05571
## 3rd Qu.:110.0    3rd Qu.:1.0000                    3rd Qu.:0.00000
## Max.   :192.0    Max.   :1.0000                    Max.   :1.00000
##      CC      Doors      Cylinders      Gears      Quarterly_Tax
##  Min.   : 1300    Min.   :2.000    Min.   :4    Min.   :3.000    Min.   :
19.00
## 1st Qu.: 1400    1st Qu.:3.000    1st Qu.:4    1st Qu.:5.000    1st Qu.:
69.00
## Median : 1600    Median :4.000    Median :4    Median :5.000    Median :
85.00
## Mean   : 1577    Mean  :4.033    Mean  :4    Mean  :5.026    Mean  :
87.12
## 3rd Qu.: 1600    3rd Qu.:5.000    3rd Qu.:4    3rd Qu.:5.000    3rd Qu.:
85.00
## Max.   :16000    Max.   :5.000    Max.   :4    Max.   :6.000    Max.
:283.00
##      Weight      Mfr_Guarantee      BOVAG_Guarantee      Guarantee_Period
##  Min.   :1000    Min.   :0.0000    Min.   :0.0000    Min.   : 3.000
## 1st Qu.:1040    1st Qu.:0.0000    1st Qu.:1.0000    1st Qu.: 3.000
## Median :1070    Median :0.0000    Median :1.0000    Median : 3.000
## Mean   :1072    Mean  :0.4095    Mean  :0.8955    Mean  : 3.815
## 3rd Qu.:1085    3rd Qu.:1.0000    3rd Qu.:1.0000    3rd Qu.: 3.000
## Max.   :1615    Max.   :1.0000    Max.   :1.0000    Max.   :36.000
##      ABS      Airbag_1      Airbag_2      Airco
##  Min.   :0.0000    Min.   :0.0000    Min.   :0.0000    Min.   :0.0000
## 1st Qu.:1.0000    1st Qu.:1.0000    1st Qu.:0.0000    1st Qu.:0.0000
## Median :1.0000    Median :1.0000    Median :1.0000    Median :1.0000
## Mean   :0.8134    Mean  :0.9708    Mean  :0.7228    Mean  :0.5084
## 3rd Qu.:1.0000    3rd Qu.:1.0000    3rd Qu.:1.0000    3rd Qu.:1.0000
## Max.   :1.0000    Max.   :1.0000    Max.   :1.0000    Max.   :1.0000
## Automatic_airco      Boardcomputer      CD_Player      Central_Lock
##  Min.   :0.00000    Min.   :0.0000    Min.   :0.0000    Min.   :0.0000
## 1st Qu.:0.00000    1st Qu.:0.0000    1st Qu.:0.0000    1st Qu.:0.0000

```

```
## Median :0.00000 Median :0.0000 Median :0.0000 Median :1.0000
## Mean :0.05641 Mean :0.2946 Mean :0.2187 Mean :0.5801
## 3rd Qu.:0.00000 3rd Qu.:1.0000 3rd Qu.:0.0000 3rd Qu.:1.0000
## Max. :1.00000 Max. :1.0000 Max. :1.0000 Max. :1.0000
## Powered_Windows Power_Steering Radio Mistlamps
## Min. :0.000 Min. :0.0000 Min. :0.0000 Min. :0.000
## 1st Qu.:0.000 1st Qu.:1.0000 1st Qu.:0.0000 1st Qu.:0.000
## Median :1.000 Median :1.0000 Median :0.0000 Median :0.000
## Mean :0.562 Mean :0.9777 Mean :0.1462 Mean :0.257
## 3rd Qu.:1.000 3rd Qu.:1.0000 3rd Qu.:0.0000 3rd Qu.:1.000
## Max. :1.000 Max. :1.0000 Max. :1.0000 Max. :1.000
## Sport_Model Backseat_Divider Metallic_Rim Radio_cassette
## Min. :0.0000 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.:0.0000 1st Qu.:1.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :0.0000 Median :1.0000 Median :0.0000 Median :0.0000
## Mean :0.3001 Mean :0.7702 Mean :0.2047 Mean :0.1455
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :1.0000 Max. :1.0000 Max. :1.0000 Max. :1.0000
## Parking_Assistant Tow_Bar
## Min. :0.000000 Min. :0.0000
## 1st Qu.:0.000000 1st Qu.:0.0000
## Median :0.000000 Median :0.0000
## Mean :0.002786 Mean :0.2779
## 3rd Qu.:0.000000 3rd Qu.:1.0000
## Max. :1.000000 Max. :1.0000
```

```
dim(Daten)
```

```
## [1] 1436 39
```

```
Daten = Daten[,-c(1:2,5:6)] #remove irrelevant info, all cars of same type etc
```

```
Daten = Daten[,-c(11)] #remove static data like cylinders, all = 4
```

```
dim(Daten)
```

```
## [1] 1436 34
```

```
head(Daten)
```

```
## Price Age_08_04 KM Fuel_Type HP Met_Color Color Automatic CC Doors
## 1 13500 23 46986 Diesel 90 1 Blue 0 2000 3
## 2 13750 23 72937 Diesel 90 1 Silver 0 2000 3
## 3 13950 24 41711 Diesel 90 1 Blue 0 2000 3
## 4 14950 26 48000 Diesel 90 0 Black 0 2000 3
## 5 13750 30 38500 Diesel 90 0 Black 0 2000 3
## 6 12950 32 61000 Diesel 90 0 White 0 2000 3
## Gears Quarterly_Tax Weight Mfr_Guarantee BOVAG_Guarantee
Guarantee_Period ABS
## 1 5 210 1165 0 1
3 1
## 2 5 210 1165 0 1
```

```

3 1
## 3 5 210 1165 1 1
3 1
## 4 5 210 1165 1 1
3 1
## 5 5 210 1170 1 1
3 1
## 6 5 210 1170 0 1
3 1
## Airbag_1 Airbag_2 Airco Automatic_airco Boardcomputer CD_Player
Central_Lock
## 1 1 1 0 0 1 0
1
## 2 1 1 1 0 1 1
1
## 3 1 1 0 0 1 0
0
## 4 1 1 0 0 1 0
0
## 5 1 1 1 0 1 0
1
## 6 1 1 1 0 1 0
1
## Powered_Windows Power_Steering Radio Mistlamps Sport_Model
Backseat_Divider
## 1 1 1 0 0 0
1
## 2 0 1 0 0 0
1
## 3 0 1 0 0 0
1
## 4 0 1 0 0 0
1
## 5 1 1 0 1 0
1
## 6 1 1 0 1 0
1
## Metallic_Rim Radio_cassette Parking_Assistant Tow_Bar
## 1 0 0 0 0
## 2 0 0 0 0
## 3 0 0 0 0
## 4 0 0 0 0
## 5 0 0 0 0
## 6 0 0 0 0

```

```
insample = data.frame(Daten[1:1000,])
```

```
#in sample regression
```

```
reg = lm(Price~., data = insample)
```

```
summary(reg, digits = 4)
```

```
##
## Call:
## lm(formula = Price ~ ., data = insample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7063.5  -731.2   -15.9    698.4   5267.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.622e+02  1.909e+03  -0.452  0.651598
## Age_08_04     -1.194e+02  4.773e+00 -25.013 < 2e-16 ***
## KM            -1.924e-02  1.586e-03 -12.133 < 2e-16 ***
## Fuel_TypeDiesel  6.742e+02  3.753e+02  1.797  0.072691 .
## Fuel_TypePetrol  1.953e+03  3.898e+02  5.009  6.50e-07 ***
## HP             2.728e+01  3.678e+00  7.417  2.64e-13 ***
## Met_Color     -1.242e+02  9.183e+01  -1.353  0.176473
## ColorBlack     8.548e+02  8.400e+02  1.018  0.309144
## ColorBlue      7.089e+02  8.393e+02  0.845  0.398566
## ColorGreen     5.538e+02  8.408e+02  0.659  0.510295
## ColorGrey      9.563e+02  8.391e+02  1.140  0.254669
## ColorRed       5.615e+02  8.393e+02  0.669  0.503674
## ColorSilver    1.012e+03  8.439e+02  1.199  0.230771
## ColorViolet    8.443e+02  1.029e+03  0.821  0.411985
## ColorWhite    -1.990e+02  8.912e+02  -0.223  0.823328
## ColorYellow    4.474e+02  1.077e+03  0.416  0.677815
## Automatic      2.188e+02  1.864e+02  1.173  0.240920
## CC            -9.369e-02  8.236e-02  -1.137  0.255615
## Doors          2.825e+01  4.444e+01  0.636  0.525022
## Gears          -8.053e+00  2.008e+02  -0.040  0.968011
## Quarterly_Tax  1.704e+01  2.045e+00  8.331  2.76e-16 ***
## Weight         1.082e+01  1.275e+00  8.491  < 2e-16 ***
## Mfr_Guarantee   2.395e+02  8.174e+01  2.930  0.003473 **
## BOVAG_Guarantee 6.628e+02  1.456e+02  4.554  5.95e-06 ***
## Guarantee_Period 5.865e+01  1.403e+01  4.182  3.16e-05 ***
## ABS            2.058e+02  1.887e+02  1.091  0.275605
## Airbag_1       -1.580e+00  3.591e+02  -0.004  0.996490
## Airbag_2        2.139e+02  1.729e+02  1.237  0.216229
## Airco          2.213e+02  1.046e+02  2.116  0.034577 *
## Automatic_airco 2.226e+03  1.896e+02  11.742 < 2e-16 ***
## Boardcomputer  -3.973e+02  1.171e+02  -3.392  0.000723 ***
## CD_Player       2.244e+02  1.012e+02  2.217  0.026869 *
## Central_Lock   -1.434e+02  1.606e+02  -0.893  0.371971
## Powered_Windows 4.556e+02  1.608e+02  2.834  0.004688 **
## Power_Steering -1.192e+02  3.872e+02  -0.308  0.758320
## Radio          5.734e+02  6.972e+02  0.822  0.411026
## Mistlamps      6.010e+01  1.219e+02  0.493  0.622003
## Sport_Model     3.644e+02  1.001e+02  3.640  0.000287 ***
## Backseat_Divider -5.531e+02  1.777e+02  -3.112  0.001912 **
## Metallic_Rim    2.127e+02  1.045e+02  2.035  0.042167 *
## Radio_cassette  -6.215e+02  6.977e+02  -0.891  0.373264
```

```
## Parking_Assistant -6.854e+02  6.045e+02  -1.134 0.257187
## Tow_Bar          -5.038e+01  9.260e+01  -0.544 0.586546
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1162 on 957 degrees of freedom
## Multiple R-squared:  0.908, Adjusted R-squared:  0.904
## F-statistic: 224.9 on 42 and 957 DF, p-value: < 2.2e-16
tab_model(reg) #Look in Viewer tab to the far right side
```

Price

Predictors

Estimates

CI

p

(Intercept)

-862.20

-4608.24 – 2883.83

0.652

Age 08 04

-119.39

-128.76 – -110.02

<0.001

KM

-0.02

-0.02 – -0.02

<0.001

Fuel Type [Diesel]

674.24

-62.18 – 1410.67

0.073

Fuel Type [Petrol]

1952.50

1187.61 – 2717.40

<0.001

HP

27.28

20.06 – 34.50

<0.001

Met Color

-124.22

-304.44 – 55.99

0.176

Color [Black]

854.75

-793.71 – 2503.21

0.309

Color [Blue]

708.85

-938.26 – 2355.96

0.399

Color [Green]

553.79

-1096.30 – 2203.88

0.510

Color [Grey]

956.34

-690.29 – 2602.97

0.255

Color [Red]

561.46

-1085.60 – 2208.52

0.504

Color [Silver]

1011.97

-644.17 – 2668.11

0.231

Color [Violet]

844.32

-1174.45 – 2863.10

0.412

Color [White]

-199.04

-1948.04 – 1549.97

0.823

Color [Yellow]

447.41

-1665.39 – 2560.21

0.678

Automatic

218.77

-147.11 – 584.65

0.241

CC

-0.09

-0.26 – 0.07

0.256

Doors

28.25

-58.95 – 115.46

0.525

Gears

-8.05

-402.02 – 385.92

0.968

Quarterly Tax

17.04

13.03 – 21.05

<0.001

Weight

10.82

8.32 – 13.33

<0.001

Mfr Guarantee

239.48

79.07 – 399.89

0.003

BOVAG Guarantee

662.81

377.17 – 948.45

<0.001

Guarantee Period

58.65

31.13 – 86.18

<0.001

ABS

205.81

-164.44 – 576.07

0.276

Airbag 1

-1.58

-706.35 – 703.19

0.996

Airbag 2

213.91

-125.33 – 553.16

0.216

Airco

221.33

16.09 – 426.58

0.035

Automatic airco

2226.31

1854.23 – 2598.39

<0.001

Boardcomputer

-397.27

-627.12 – -167.41

0.001

CD Player

224.39

25.75 – 423.04

0.027

Central Lock

-143.44

-458.59 – 171.71

0.372

Powered Windows

455.64

140.17 – 771.11

0.005

Power Steering

-119.18

-879.10 – 640.74

0.758

Radio

573.41

-794.80 – 1941.61

0.411

Mistlamps

60.10

-179.06 – 299.27

0.622

Sport Model

364.38

167.93 – 560.83

<0.001

Backseat Divider

-553.09

-901.85 – -204.33

0.002

Metallic Rim

212.67

7.54 – 417.80

0.042

Radio cassette

-621.52

-1990.74 – 747.71

0.373

Parking Assistant

-685.38

-1871.74 – 500.98

0.257

Tow Bar

-50.38

-232.09 – 131.34

0.587

Observations

1000

R2 / R2 adjusted

0.908 / 0.904

```
# Diagnostics of one specific regression
```

```
##Fehlspezifikation-----
```

```
resettest(reg)#H0 No spec error
```

```
##
```

```
## RESET test
```

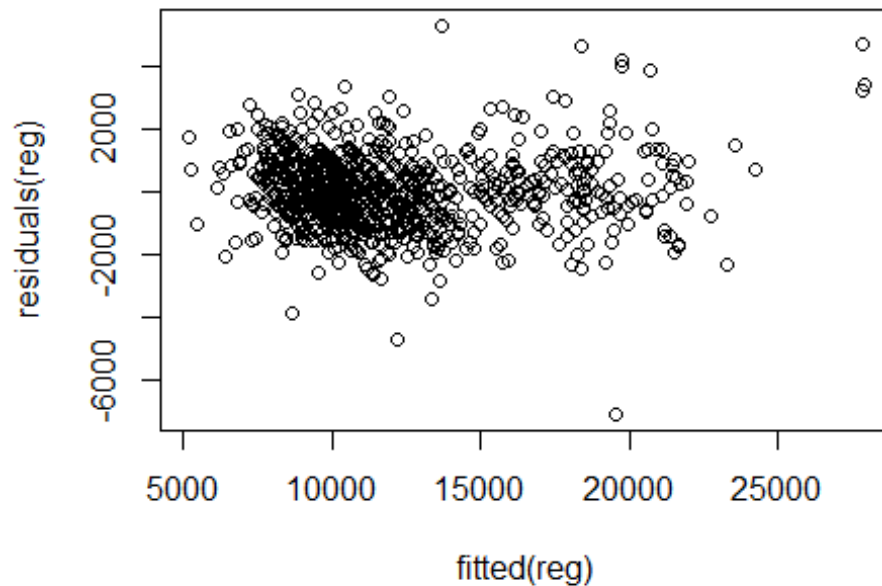
```
##
```

```
## data: reg
```

```
## RESET = 65.109, df1 = 2, df2 = 955, p-value < 2.2e-16
```

```
##Hetero-----
```

```
plot(fitted(reg),residuals(reg)) #, type='l')
```

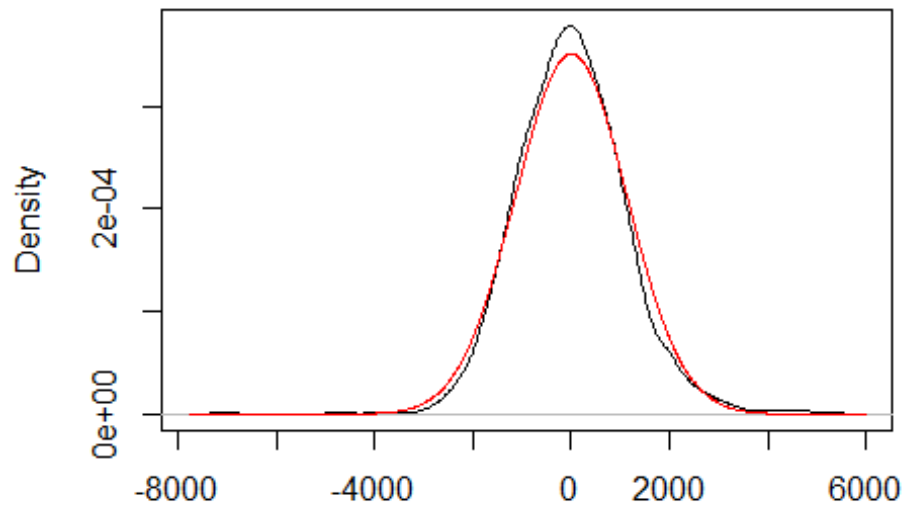


```
#Breusch-Pagan-Test auf Heteroskedastizität
bptest(reg)#H0: Homoscedasticity! but assumes normal resis

##
## studentized Breusch-Pagan test
##
## data: reg
## BP = 284.47, df = 42, p-value < 2.2e-16

#Normality-----
kernel_den = density(residuals(reg))
plot(kernel_den, main="Smoothed Residual Distribution", xlab = " ")
x = seq(min(kernel_den$x), max(kernel_den$x), length = 1000)
#compare to normal
std_den = dnorm(x, mean = mean(residuals(reg)), sd = sd(residuals(reg)))
lines(x, std_den, type="l", col="red")
```

Smoothed Residual Distribution



```
#as test
shapiro.test(residuals(reg))#H0: Is normal

##
##  Shapiro-Wilk normality test
##
## data:  residuals(reg)
## W = 0.97645, p-value = 1.181e-11

#NOT nORMALLLLLLLLLLLLLL

##conclusion-----
#Hence Newey-West again!!!
summary(reg)

##
## Call:
## lm(formula = Price ~ ., data = insample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7063.5  -731.2   -15.9    698.4   5267.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -8.622e+02  1.909e+03  -0.452  0.651598
## Age_08_04     -1.194e+02  4.773e+00 -25.013 < 2e-16 ***
## KM            -1.924e-02  1.586e-03 -12.133 < 2e-16 ***
```

```

## Fuel_TypeDiesel      6.742e+02  3.753e+02   1.797 0.072691 .
## Fuel_TypePetrol      1.953e+03  3.898e+02   5.009 6.50e-07 ***
## HP                   2.728e+01  3.678e+00   7.417 2.64e-13 ***
## Met_Color            -1.242e+02  9.183e+01  -1.353 0.176473
## ColorBlack           8.548e+02  8.400e+02   1.018 0.309144
## ColorBlue            7.089e+02  8.393e+02   0.845 0.398566
## ColorGreen           5.538e+02  8.408e+02   0.659 0.510295
## ColorGrey            9.563e+02  8.391e+02   1.140 0.254669
## ColorRed             5.615e+02  8.393e+02   0.669 0.503674
## ColorSilver          1.012e+03  8.439e+02   1.199 0.230771
## ColorViolet          8.443e+02  1.029e+03   0.821 0.411985
## ColorWhite          -1.990e+02  8.912e+02  -0.223 0.823328
## ColorYellow          4.474e+02  1.077e+03   0.416 0.677815
## Automatic            2.188e+02  1.864e+02   1.173 0.240920
## CC                   -9.369e-02  8.236e-02  -1.137 0.255615
## Doors                2.825e+01  4.444e+01   0.636 0.525022
## Gears                -8.053e+00  2.008e+02  -0.040 0.968011
## Quarterly_Tax        1.704e+01  2.045e+00   8.331 2.76e-16 ***
## Weight               1.082e+01  1.275e+00   8.491 < 2e-16 ***
## Mfr_Guarantee         2.395e+02  8.174e+01   2.930 0.003473 **
## BOVAG_Guarantee       6.628e+02  1.456e+02   4.554 5.95e-06 ***
## Guarantee_Period      5.865e+01  1.403e+01   4.182 3.16e-05 ***
## ABS                  2.058e+02  1.887e+02   1.091 0.275605
## Airbag_1             -1.580e+00  3.591e+02  -0.004 0.996490
## Airbag_2             2.139e+02  1.729e+02   1.237 0.216229
## Airco                2.213e+02  1.046e+02   2.116 0.034577 *
## Automatic_airco       2.226e+03  1.896e+02  11.742 < 2e-16 ***
## Boardcomputer        -3.973e+02  1.171e+02  -3.392 0.000723 ***
## CD_Player            2.244e+02  1.012e+02   2.217 0.026869 *
## Central_Lock          -1.434e+02  1.606e+02  -0.893 0.371971
## Powered_Windows       4.556e+02  1.608e+02   2.834 0.004688 **
## Power_Steering        -1.192e+02  3.872e+02  -0.308 0.758320
## Radio                 5.734e+02  6.972e+02   0.822 0.411026
## Mistlamps            6.010e+01  1.219e+02   0.493 0.622003
## Sport_Model           3.644e+02  1.001e+02   3.640 0.000287 ***
## Backseat_Divider     -5.531e+02  1.777e+02  -3.112 0.001912 **
## Metallic_Rim          2.127e+02  1.045e+02   2.035 0.042167 *
## Radio_cassette        -6.215e+02  6.977e+02  -0.891 0.373264
## Parking_Assistant     -6.854e+02  6.045e+02  -1.134 0.257187
## Tow_Bar              -5.038e+01  9.260e+01  -0.544 0.586546
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1162 on 957 degrees of freedom
## Multiple R-squared:  0.908, Adjusted R-squared:  0.904
## F-statistic: 224.9 on 42 and 957 DF, p-value: < 2.2e-16

```

#75% is idiosyncratic risk!

`coeftest(reg,vcov=NeweyWest(reg))` *#now use heteroskedasticity and autocorrelation*

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.6220e+02 5.1066e+03 -0.1688 0.8659582
## Age_08_04 -1.1939e+02 5.8701e+00 -20.3387 < 2.2e-16 ***
## KM -1.9238e-02 2.1295e-03 -9.0339 < 2.2e-16 ***
## Fuel_TypeDiesel 6.7424e+02 4.6166e+02 1.4605 0.1444828
## Fuel_TypePetrol 1.9525e+03 6.4231e+02 3.0398 0.0024314 **
## HP 2.7279e+01 7.2983e+00 3.7377 0.0001967 ***
## Met_Color -1.2422e+02 8.7557e+01 -1.4187 0.1563014
## ColorBlack 8.5475e+02 5.4789e+02 1.5601 0.1190740
## ColorBlue 7.0885e+02 5.5157e+02 1.2851 0.1990513
## ColorGreen 5.5379e+02 5.4853e+02 1.0096 0.3129463
## ColorGrey 9.5634e+02 5.4859e+02 1.7433 0.0816049 .
## ColorRed 5.6146e+02 5.4476e+02 1.0307 0.3029647
## ColorSilver 1.0120e+03 5.5893e+02 1.8105 0.0705250 .
## ColorViolet 8.4432e+02 6.3439e+02 1.3309 0.1835346
## ColorWhite -1.9904e+02 6.6600e+02 -0.2989 0.7651160
## ColorYellow 4.4741e+02 6.7922e+02 0.6587 0.5102364
## Automatic 2.1877e+02 2.2506e+02 0.9721 0.3312552
## CC -9.3686e-02 7.0230e-02 -1.3340 0.1825239
## Doors 2.8255e+01 6.9440e+01 0.4069 0.6841760
## Gears -8.0530e+00 2.1839e+02 -0.0369 0.9705930
## Quarterly_Tax 1.7039e+01 4.0334e+00 4.2246 2.623e-05 ***
## Weight 1.0824e+01 4.8790e+00 2.2185 0.0267523 *
## Mfr_Guarantee 2.3948e+02 7.2689e+01 3.2946 0.0010218 **
## BOVAG_Guarantee 6.6281e+02 1.5729e+02 4.2140 2.746e-05 ***
## Guarantee_Period 5.8653e+01 1.8195e+01 3.2235 0.0013090 **
## ABS 2.0581e+02 2.0917e+02 0.9839 0.3253941
## Airbag_1 -1.5801e+00 3.6708e+02 -0.0043 0.9965663
## Airbag_2 2.1391e+02 1.9105e+02 1.1197 0.2631294
## Airco 2.2133e+02 1.0205e+02 2.1689 0.0303326 *
## Automatic_airco 2.2263e+03 2.8627e+02 7.7771 1.914e-14 ***
## Boardcomputer -3.9727e+02 1.2064e+02 -3.2929 0.0010278 **
## CD_Player 2.2439e+02 9.3562e+01 2.3983 0.0166603 *
## Central_Lock -1.4344e+02 1.5728e+02 -0.9120 0.3620150
## Powered_Windows 4.5564e+02 1.6477e+02 2.7653 0.0057957 **
## Power_Steering -1.1918e+02 3.6908e+02 -0.3229 0.7468302
## Radio 5.7341e+02 5.7316e+02 1.0004 0.3173533
## Mistlamps 6.0104e+01 1.3057e+02 0.4603 0.6454021
## Sport_Model 3.6438e+02 1.2466e+02 2.9229 0.0035491 **
## Backseat_Divider -5.5309e+02 1.7348e+02 -3.1882 0.0014781 **
## Metallic_Rim 2.1267e+02 1.1695e+02 1.8185 0.0693003 .
## Radio_cassette -6.2152e+02 5.6410e+02 -1.1018 0.2708340
## Parking_Assistant -6.8538e+02 3.0938e+02 -2.2154 0.0269695 *
## Tow_Bar -5.0375e+01 9.0803e+01 -0.5548 0.5791786
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


#consistent (HAC) covariance matrix estimators. Note the decreased t value!

vif(reg)[,1]#could reduce X dim ...

##	Age_08_04	KM	Fuel_Type	HP
##	4.700974	2.333795	9.783617	2.427803
##	Met_Color	Color	Automatic	CC
##	1.352826	2.196029	1.176867	1.216384
##	Doors	Gears	Quarterly_Tax	Weight
##	1.328536	1.394619	5.882953	4.063355
##	Mfr_Guarantee	BOVAG_Guarantee	Guarantee_Period	ABS
##	1.227178	1.462688	1.696901	1.510669
##	Airbag_1	Airbag_2	Airco	Automatic_airco
##	1.781138	3.065307	1.931547	1.915450
##	Boardcomputer	CD_Player	Central_Lock	Powered_Windows
##	2.480866	1.566257	4.458877	4.537778
##	Power_Steering	Radio	Mistlamps	Sport_Model
##	1.856590	47.912306	2.336991	1.522498
##	Backseat_Divider	Metallic_Rim	Radio_cassette	Parking_Assistant
##	3.772178	1.407131	47.736829	1.078769
##	Tow_Bar			
##	1.175081			

#end